



grey: the insert's seam

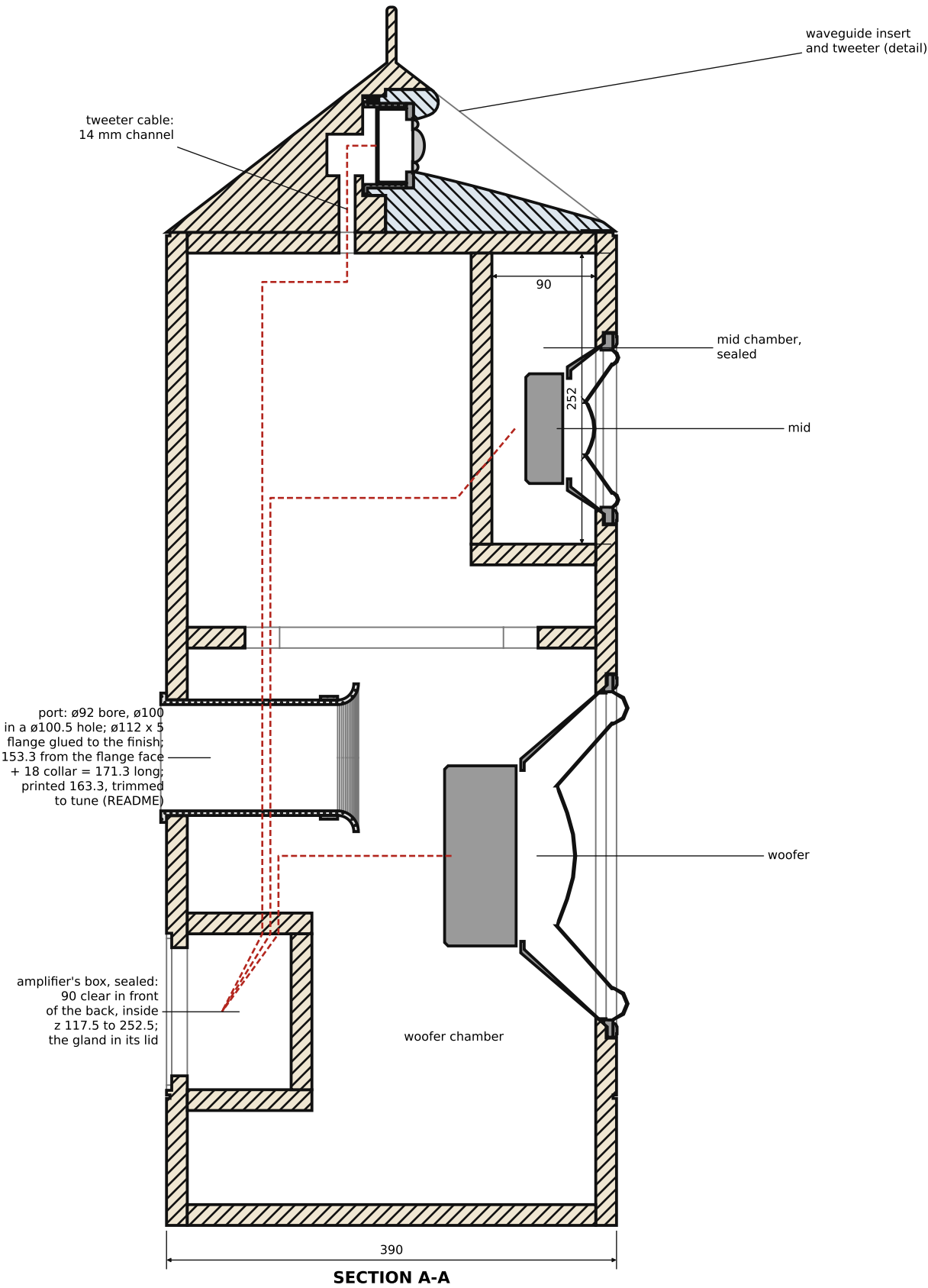
Note

5. Finish: colour coat and 2K clear; the Facts printed between them.

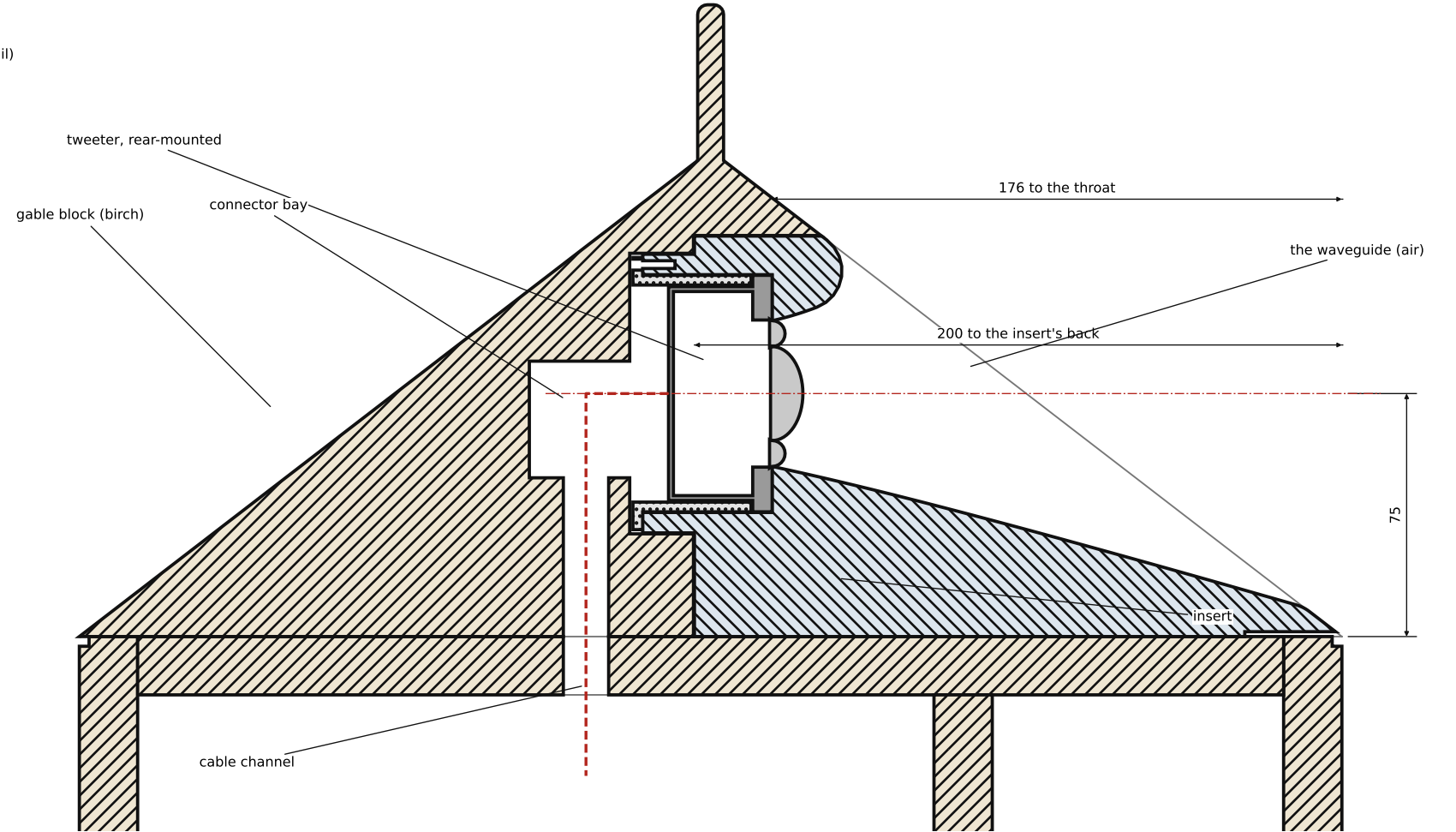
mm | issue 2026-10-09 | from fab/sheets.py (the CAD) | fits as dimensioned

1 of 6

EM-FS-001 rev C



red dashed: the cables' runs to the amplifier (sheet 4)



DETAIL: THE ROOF, 1:2

The roof

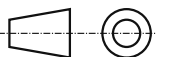
1. The waveguide insert (light blue) sits in a pocket in the gable block, 0.3 mm clear all round: the pocket $+0.2/0$, the insert $0/-0.15$, so 0.3 to 0.65 a side. Paint only the insert's face: mask its sides, base and back and the pocket's walls. It slides out forward, level, with the tweeter on it.
2. The tweeter is rear-mounted: it goes in from behind through a $\varnothing 73.4$ bore and its front ring seats on the ring round the throat, its dome and surround filling the 45 mm throat, so the wall runs on from the surround with no step. A printed sleeve over its motor holds it there, screwed to the boss's back face (sheet 3).
3. Its lead leaves the boss's open back into the connector bay behind it ($\varnothing 36 \times 35$ deep), where it plugs into the cabinet's lead; that runs down a 14 mm channel through the block and the top panel, sealed round the cable with neutral-cure silicone from the bay.
4. Four magnets and two pins in the insert's back face meet their partners in the pocket's back wall (sheet 3).

earmilk floorstander | Section on the centreline

Section A-A on the centre plane (x 195), from the left, at 1:5; the roof at 1:2. The insert hatched the other way.

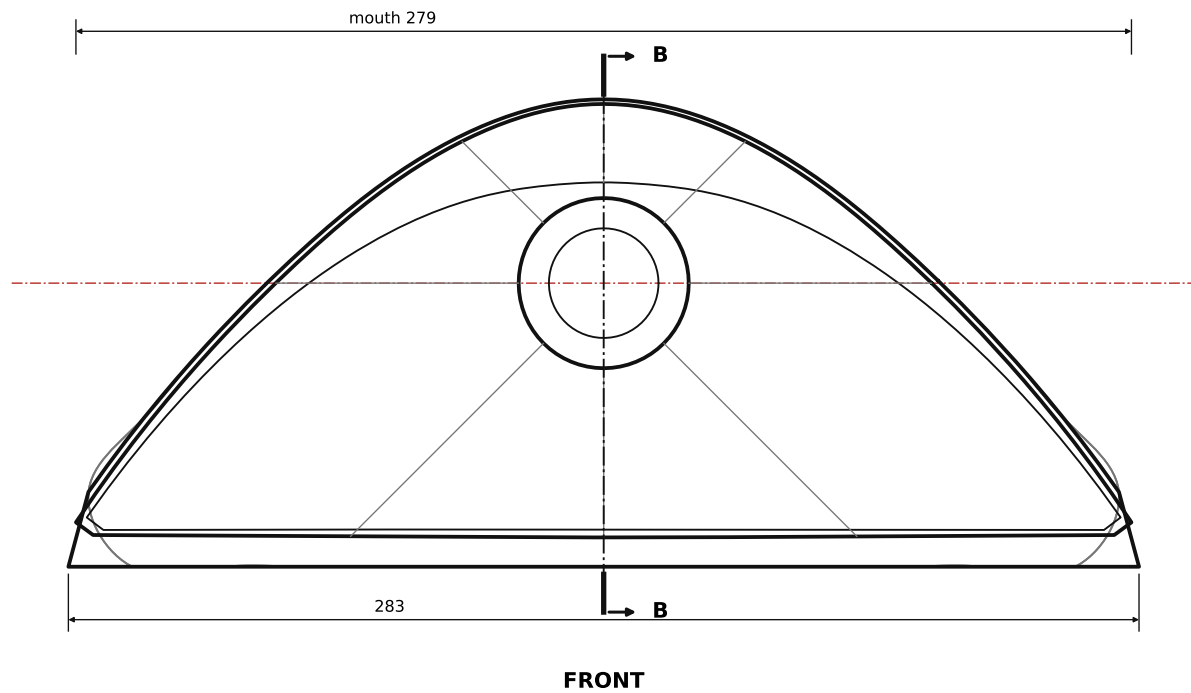
mm | issue 2026-10-09 | from fab/sheets.py (the CAD) | fits as dimensioned

Tolerances: panels ± 0.5 , pocket depths ± 0.2 , holes $+0.2/-0$, prints ± 0.15 . Check bought parts first.



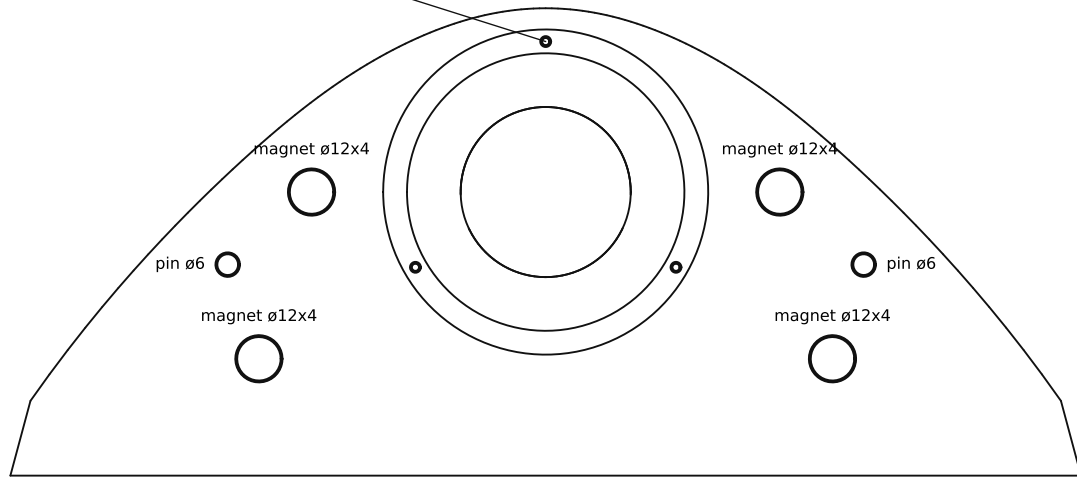
2 of 6

EM-FS-002 rev C

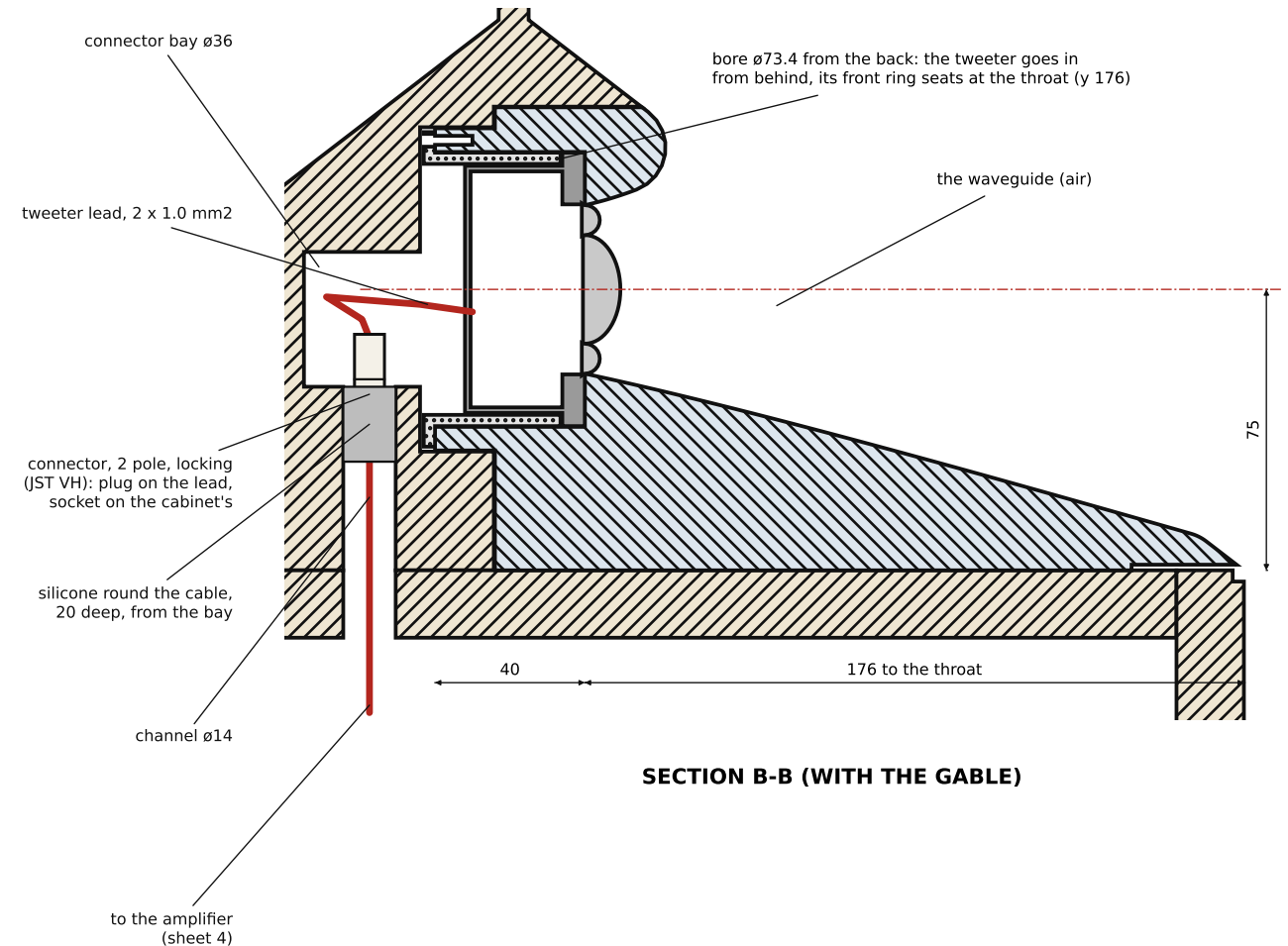


FRONT

3 x $\varnothing 2.4 \times 10$ pilots on $\varnothing 79.6$ in the boss's back face, one at the top, 120° apart, for the retaining sleeve's 3.0 x 12 thread-forming screws for plastics (WN 1411 / PT K30, pan head)



BACK (THE FACE THAT MEETS THE POCKET)



SECTION B-B (WITH THE GABLE)

Fitting the tweeter and the insert

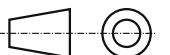
1. Solder the tweeter's lead (2 x 1.0 mm2, 120 mm) to its tabs and crimp the plug half of a two-pole locking connector (JST VH) on its end.
2. Pass the tweeter in from behind through the $\varnothing 73.4$ bore, dome first, and seat its front ring on a 0.5 mm foam gasket on the throat's seat. Slide the printed retaining sleeve (stl/tweeter-retainer.stl: bore $\varnothing 67$, outside $\varnothing 73$, flange $\varnothing 84 \times 3$) over its motor, lead through it, and drive its 3 3.0 x 12 thread-forming screws for plastics (WN 1411 / PT K30, pan head) into the boss until the ring is held: no thread to SB's own screw holes is needed. Measure the motor first: the sleeve's bore is its diameter + 1.
3. Glue the magnets into the insert's back with epoxy, polarity marked, and their partners into the pocket's back wall, opposite poles out. Bond the two pins into the insert with epoxy.
4. The cabinet's lead ends in the socket 250 mm above the top panel's underside. With the insert out, draw the socket forward out of the bay through the empty pocket, plug the tweeter's lead into it, and tuck both and the slack back into the bay. Then seal the channel round the cable with 20 mm of neutral-cure silicone from the bay.
5. Slide the insert in, level, until the pins seat and the magnets pull it home: its face flush with the roof, the seam even. Fit: the pocket +0.2/0, the insert 0/-0.15 (0.3 to 0.65 a side); paint only the insert's face.
6. Service: pull it out by a ribbon loop glued in the 12 x 1.5 x 30 groove under its front edge, unplug, and the tweeter comes out with it.
7. The pocket (gable-block.step): the insert's outline + 0.3 a side (283.9 wide at its base), from the slope to its back wall at y 200.3; the boss's bore $\varnothing 86.6$ to y 220; the connector bay $\varnothing 36$ from y 216 to 251 on an axis at z 927; the channel $\varnothing 14$ at x 195, y 233.5 down to the top panel; internal corners R3 or less (the insert's match); dowel holes $\varnothing 10 \times 20$ in the base at (60, 215), (330, 215), (60, 330), (330, 330).
8. Fixings, from the centreline and the insert's base (mirrored in the pocket's wall): magnets ± 75.9 across, 30.9 up; ± 61.9 across, 75.0 up; pins ± 84.1 across, 55.8 up. Magnets $\varnothing 12 \times 4$ in $\varnothing 12.2 \times 4.3$ holes, epoxied, poles marked; pins $\varnothing 6 \times 10$ bonded with epoxy in $\varnothing 6.1 \times 5.5$ in the insert, a slip fit in $\varnothing 6.2 \times 5.5$ in the wall (these holes are outside the title block's +0.2/-0).

earmilk floorstander | Waveguide insert and tweeter mount

The insert at 1:2: front, section on the axis, back; the pocket; fitting and service.

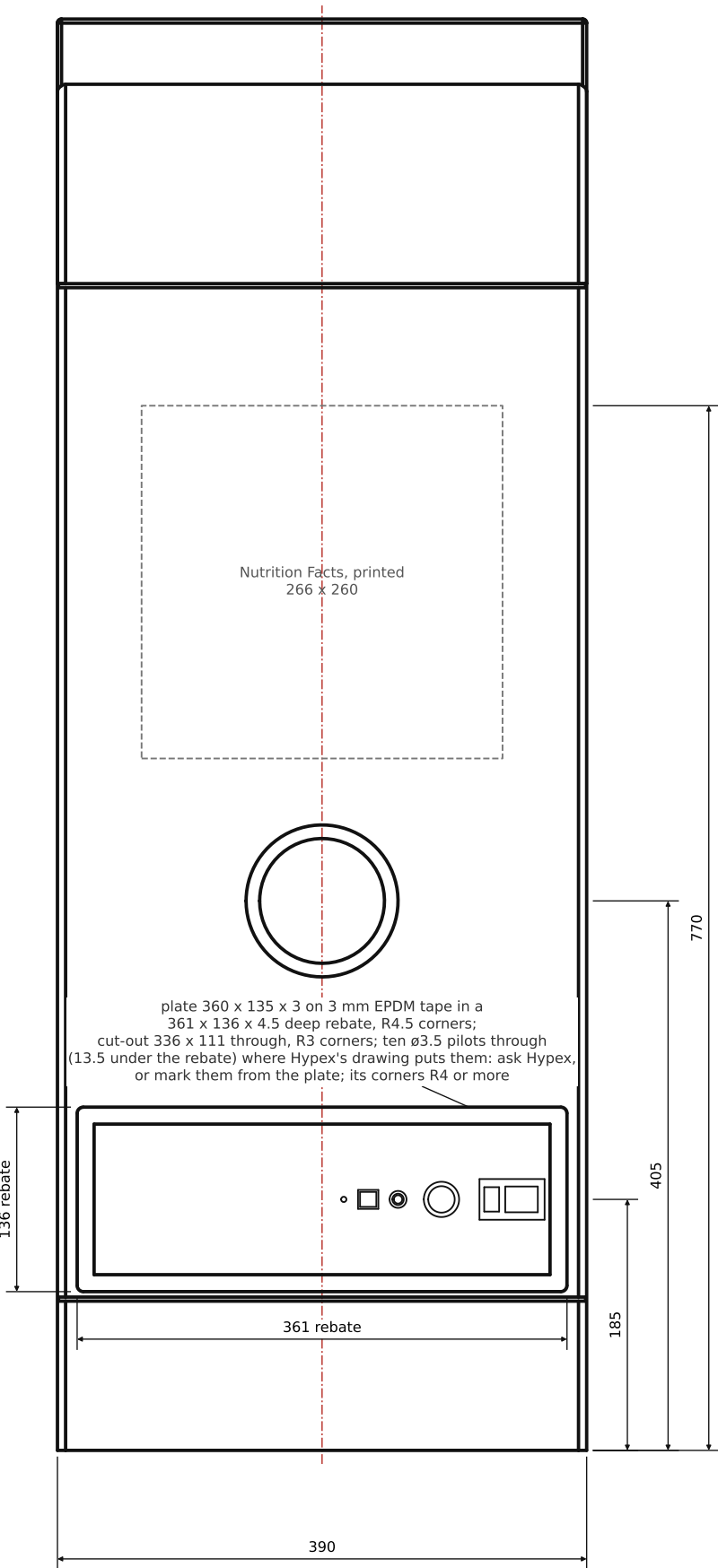
mm | issue 2026-10-09 | from fab/sheets.py (the CAD) | fits as dimensioned

Tolerances: panels ± 0.5 , pocket depths ± 0.2 , holes +0.2/-0, prints ± 0.15 . Check bought parts first.

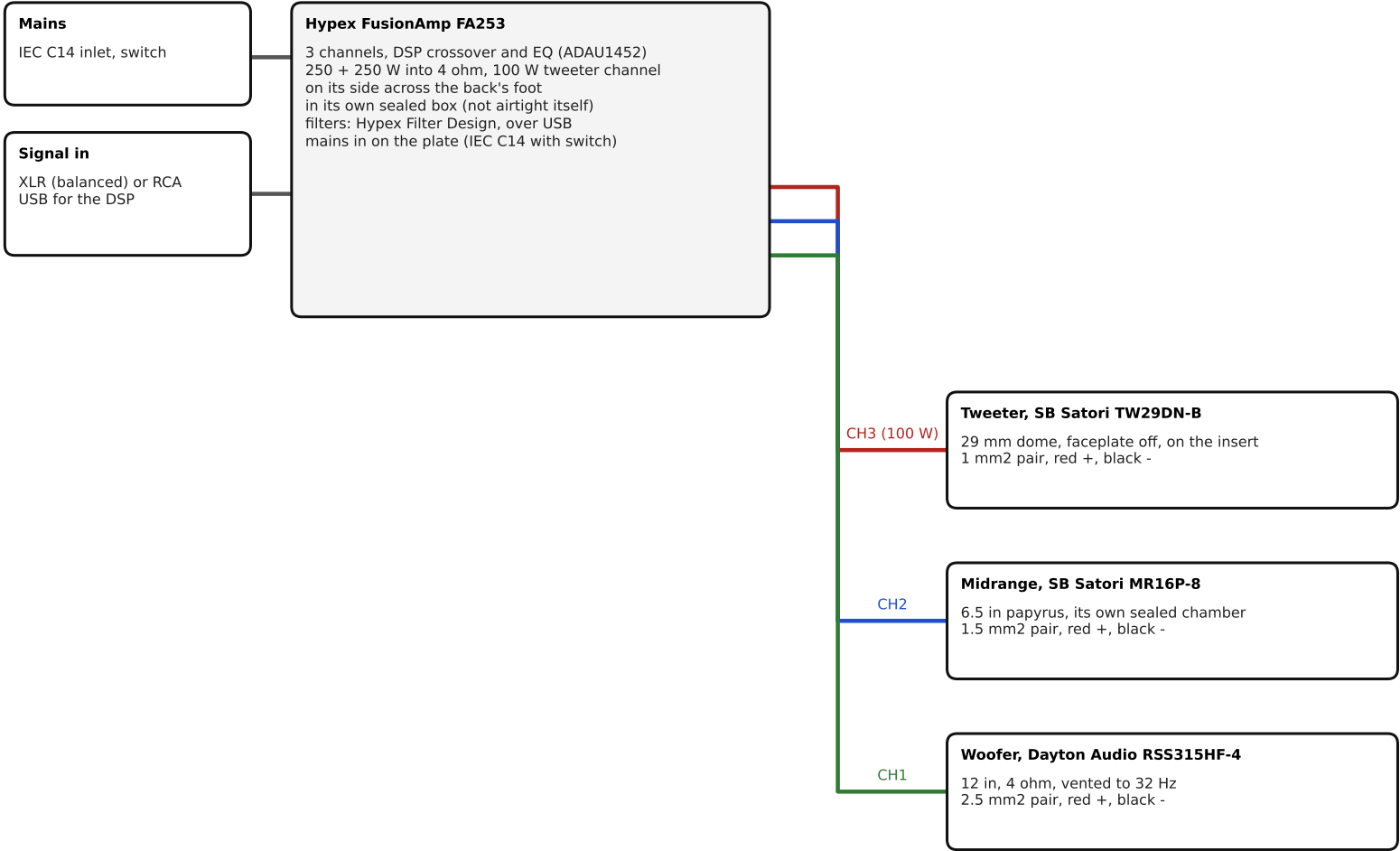


3 of 6

EM-FS-003 rev C



BACK



Runs, connectors and seals

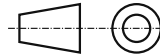
1. Woofer (CH1): from the amplifier box's gland straight to the woofer's terminals, 0.6 m of 2.5 mm2. Fit 6.3 mm push-on terminals to suit the driver.
2. Mid (CH2): from the gland up through the brace's window, 30 mm in from its back edge, then through the 12 mm hole in the mid chamber's divider (x 75, z 630), 1.0 m of 1.5 mm2. Seal the hole round the cable with putty: the mid's chamber must stay closed.
3. Tweeter (CH3): from the gland up through the brace's window to the top panel, through the 14 mm channel in the top panel and the gable block, 1.3 m of 1.0 mm2, ending in the socket of a two-pole locking connector (JST VH) in the bay behind the insert. The supplied 1.25 m Hypex harness is too short: extend it (note 5).
4. Seal the tweeter cable in its channel with 20 mm of neutral-cure silicone from the bay (sheet 3). The insert's pocket is open to the room through its 0.3 mm seam, so an unsealed channel would be a leak in the woofer's box.
5. Round-sheathed cable throughout (H05VV-F 2 x 2.5, 2 x 1.5 and 2 x 1.0 mm2, about 9.0, 8.0 and 6.4 across), so the gland's three-hole seal grips it; splice the tweeter's extension to Hypex's harness inside the amplifier's box (crimped butt splices under heat-shrink).
6. The amplifier box's lid: an M25 x 1.5 long-thread (15 mm) gland with a three-hole seal for the three pairs, in a ø25.5 hole 314 from the left side's inner face and 43 from the lid's front edge, its lock nut in a ø40 x 8 counterbore under the lid; seal it. The box itself is glued and sealed; its front is the woofer chamber's wall.
7. Polarity, delay and the crossover (about 2.8 kHz tweeter to mid, from the waveguide study) are set in the DSP: wire red to + throughout and let the filters do the rest.

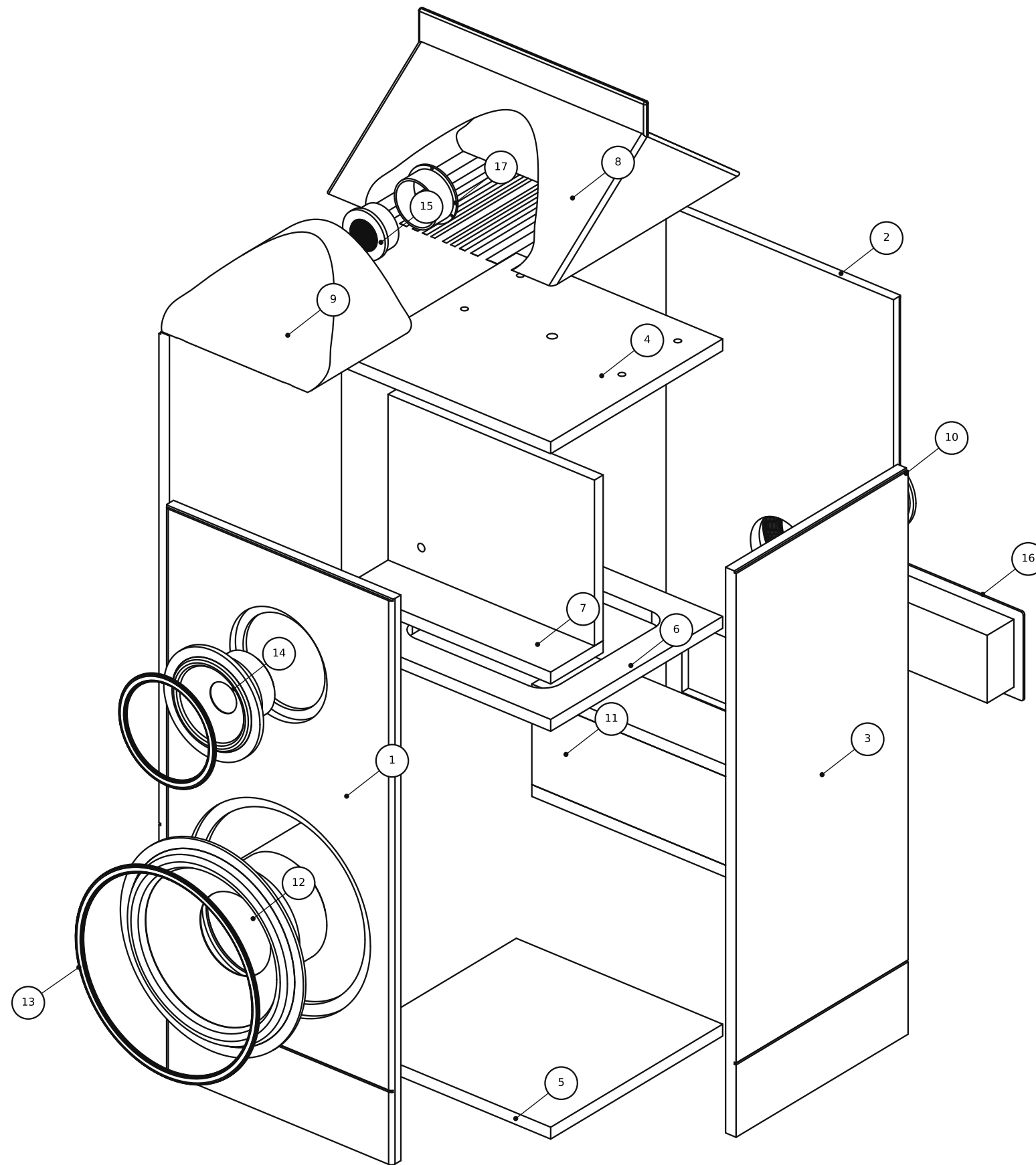
earmilk floorstander | Back and wiring

The back at 1:5 and the amplifier, drivers, cables and connectors. Active: the amplifier's DSP is the crossover.

mm | issue 2026-10-09 | from fab/sheets.py (the CAD) | fits as dimensioned

Tolerances: panels ±0.5, pocket depths ±0.2, holes +0.2/-0, prints ±0.15. Check bought parts first.





Parts

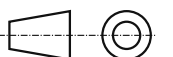
- ① **Front baffle x1**
18 mm Baltic birch, CNC: cutouts and rebates for the drivers
- ② **Back x1**
18 mm birch: port hole, the amplifier's cutout and rebate
- ③ **Sides x2**
18 mm birch
- ④ **Top panel x1**
18 mm birch, the tweeter cable's 14 mm hole, four dowel holes
- ⑤ **Bottom panel x1**
18 mm birch
- ⑥ **Window brace x1**
18 mm birch
- ⑦ **Mid chamber shelf and divider x2**
18 mm birch; the divider's 12 mm cable hole
- ⑧ **Gable block x1**
laminated birch, CNC: the insert's pocket and the cable channel
- ⑨ **Waveguide insert x1**
CNC from solid (MDF, HDPE) or printed (SLA, MJF); magnets, pins
- ⑩ **Port x1**
printed tube and flare collar, 92 bore
- ⑪ **Amplifier box x3**
18 mm birch floor, lid (gland) and front, glued and sealed
- ⑫ **Woofer x1**
Dayton Audio RSS315HF-4, 12 in, 4 ohm, black anodised aluminium cone
- ⑬ **Trim rings x2**
printed, satin black or gunmetal, over each frame and its screws
- ⑭ **Midrange x1**
SB Acoustics Satori MR16P-8, 6.5 in papyrus cone
- ⑮ **Tweeter x1**
SB Acoustics Satori TW29DN-B, faceplate off, rear-mounted on the insert
- ⑯ **Amplifier x1**
Hypex FusionAmp FA253, DSP, 3 channels
- ⑰ **Tweeter retainer x1**
printed sleeve over the motor, screwed to the boss: 3 x 3.0 x 12 PT screws

earmilk floorstander | Exploded view and parts

Isometric, not to scale: the parts pulled apart along the way they go together, and what each one is.

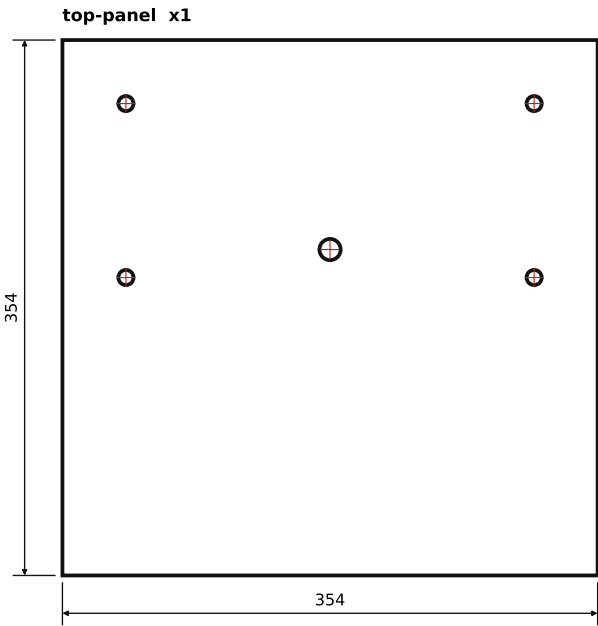
mm | issue 2026-10-09 | from fab/sheets.py (the CAD) | fits as dimensioned

Tolerances: panels ± 0.5 , pocket depths ± 0.2 , holes $+0.2/-0$, prints ± 0.15 . Check bought parts first.

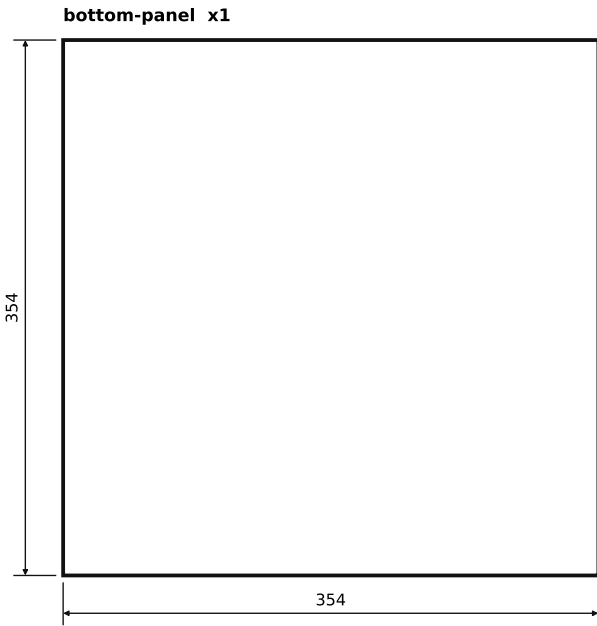


5 of 6

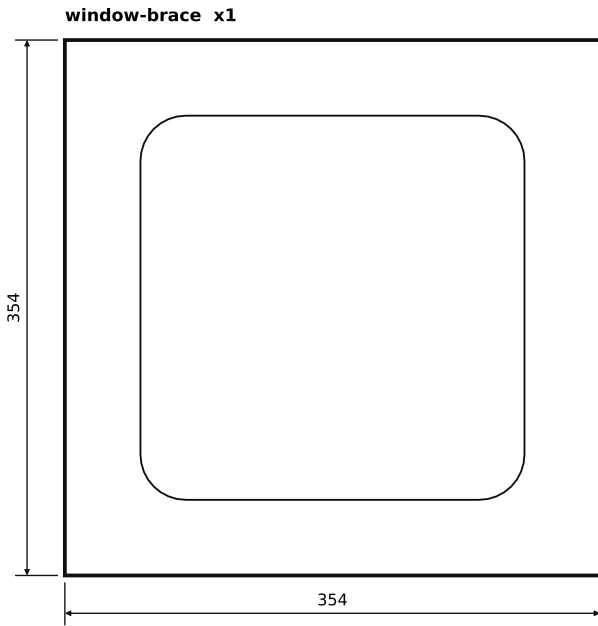
EM-FS-005 rev C



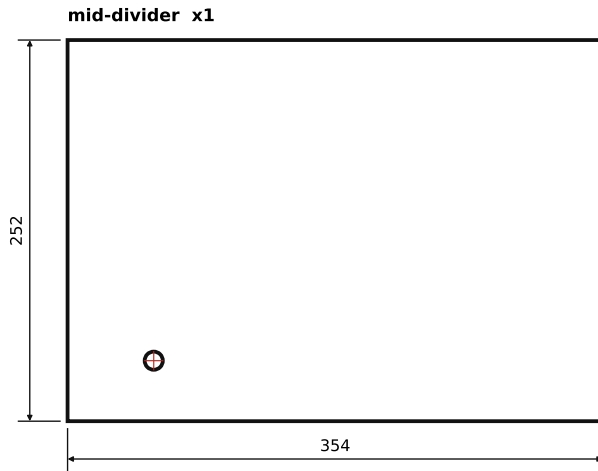
ø14 at (177, 215.5): cut inside
ø10 at (42, 197): drill ø10, 10 deep
ø10 at (312, 197): drill ø10, 10 deep
ø10 at (42, 312): drill ø10, 10 deep
ø10 at (312, 312): drill ø10, 10 deep
upper face up (front edge at the bottom of the drawing); dowels register the gable block; the tweeter cable's hole is sealed with silicone from the bay after wiring



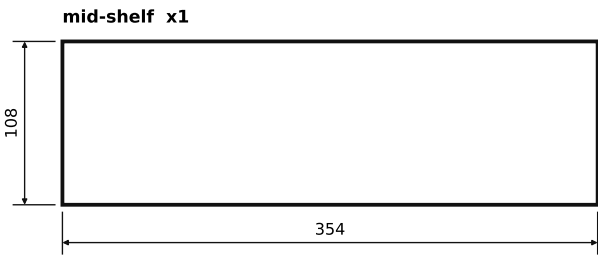
either face



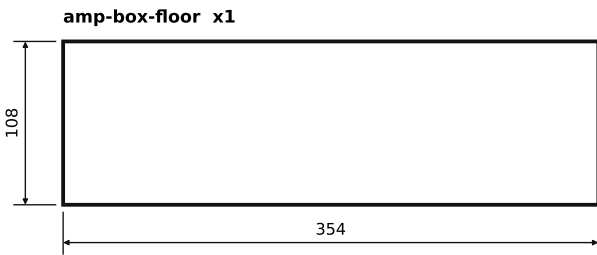
cut inside: 50 to 304 across, 50 to 304 up
sits at z 500 to 518, glued to all four walls



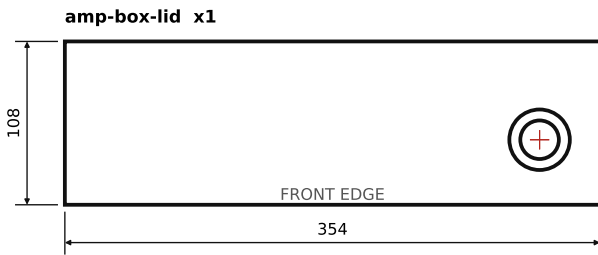
ø12 at (57, 40): cut inside
y 108 to 126, z 590 to 842; seal the wire hole after wiring



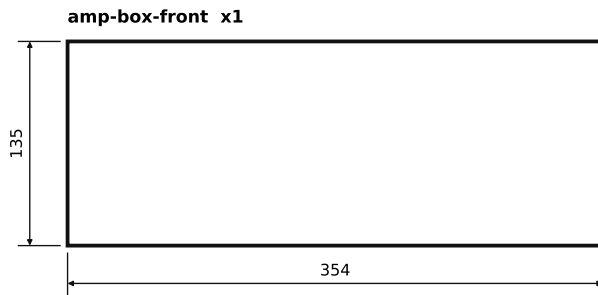
z 572 to 590, from the baffle back to under the divider



the amplifier box's floor, z 99.5 to 117.5, against the back



ø25.5 at (314, 43): cut inside
ø40 at (314, 43): pocket 8 mm deep from the underside, concentric with the gland's hole (cut the hole, turn the panel over, centre on it; before the lid goes in)
the amplifier box's lid, z 252.5 to 270.5, its front edge (the bottom of the drawing) over the box's front; the gland's ø25.5 hole 314 from the left, 43 from the front edge, its nut in a ø40 x 8 counterbore from the underside; sealed after wiring



the amplifier box's front, between floor and lid, 90 in front of the back's inner face; glue and seal all round

earmilk floorstander | Panel details

At 1:5: each panel seen from its outer (upper) face, from its lower-left corner as its DXF; holes from that corner. 18 mm birch.

mm | issue 2026-10-09 | from fab/sheets.py (the CAD) | fits as dimensioned

Tolerances: panels ±0.5, pocket depths ±0.2, holes +0.2/-0, prints ±0.15. Check bought parts first.

